Hang Lu

PhD candidate in Electrical Engineering
Optoelectronics and Photonics

Saudi Arabia 23955
King Abdullah University of Science and Technology
Photonics Lab
hang.lu@kaust.edu.sa

Education

King Abdullah University of Science and Technology (KAUST) **2021.09—Present**
PhD student in Electrical and Computer Engineering Advisor: Prof. Boon S. Ooi

King Abdullah University of Science and Technology (KAUST) **2021.09—2022.12**
Master in Electrical and Computer Engineering Advisor: Prof. Boon S. Ooi

University of Electronic Science and Technology of China (UESTC) **2017.09—2021.06**
Bachelor in Electronic and Information Engineering

Research

Semiconductor optoelectronics and photonic integrated circuit (PIC) with hands-on experience across components design (laser, microLED, waveguide, grating, and PD), epitaxy/device simulation, nanofabrication, characterization, and photonic integration. Built full-stack PIC development workflows from device physics modeling → fabrication → optical/electrical test → performance analysis. Evaluated the device performance for high-speed communication, beam steering, and imaging using customized optical and RF measurement setups.

Core skills

- **Photonic Integrated Circuit & System Architecture:** Developed emitter-detector integrated systems enabling compact high-speed electro-optical operation. Designed photonic integrated circuit (PIC) for high-speed 2D beam-steering architectures, involving lasers, couplers, modulators, waveguides, and gratings co-design, system-level performance evaluation, and system-level trade-off analysis.
- **Optoelectronic Design and Engineering:** 5 years of experience in designing, optimizing, and characterizing lasers, micro-LEDs, and PDs for high-speed modulation. Hands-on experience analyzing III-V epitaxial structures. Proficient in COMSOL, Lumerical, Tidy3D, R-Soft, Silvaco for III–V device physics and electro-optical modeling.
- **High-Speed Optical Communication:** 5 years of hands-on experience with GHz-class optical communication links (fiber and free-space), including link bring-up, alignment, coupling, and system validation. Skilled with VNA, AWG, BERT, high-speed oscilloscopes, and PD/TIA chains for S-parameters, bandwidth, eye-diagram characterization, jitter analysis, and parasitic extraction.
- **Nanofabrication & Process Engineering:** 5 years hands-on cleanroom experience: photolithography, ICP-RIE, e-beam evaporation, ALE, sputtering, PECVD, wet etching, thermal oxidation, and wire bonding. Skilled in translating device designs into fabricated, functional prototypes through full-stack process integration.
- **Optical System Design & Prototyping:** Designed and built multiple custom optical benches for high-speed communication, imaging, display, and coherence characterization. Expertise in free-space optics, alignment tolerances, speckle analysis, and system-level optical debugging.
- **Signal processing and programming:** PyTorch (for machine learning), LabView, Python, MATLAB, Origin.
- **Optical Metrology:** Expertise in SEM, photoluminescence (PL) and UV-Vis spectroscopy.
- **Machine Learning**: Supervised learning for 2D/3D signal classification task.

Professional experience

Internship at Meta (2025.06—2025.11)

- High-speed photonic integrated circuit (PIC) modeling and architecture design
  - Designed and optimized PIC-based architectures for high-speed 2D beam steering, including modulator–waveguide–grating co-design and display performance evaluation.
  - Performed chip-scale and free-space optical design to integrate PIC outputs with system-level optics.
  - Developed automated MATLAB/Python tools for optical modeling, system analysis, and performance prediction.
  - Collaborated with cross-functional teams (optics, electronics, mechanics) to define specifications and validate early prototypes.

MS and PhD at KAUST (2021.09—now):

- **Led design and prototyping of novel chaotic-cavity laser systems for high-speed communication and illumination applications.**
  - Designed and simulated chaotic-cavity VCSELs using COMSOL; fabricated devices via nanofabrication processes.
  - Built test platforms and characterized IV/LIV, modulation response, RF spectra, coherence, far-field/near-field, and thermal behavior. Applied PL spectroscopy, UV-Vis, SEM, and EDX for material and device analysis.
  - Constructed fiber and free-space GHz communication links, validating bandwidth, eye diagrams, parasitics, and system stability. Applied OOK and OFDM for the communication test.
  - Built optical benches from scratch to correlate optical modeling with experimental results.
- **Led development of integrated optoelectronic chips for secure communication, random bit generation, and physical unclonable functions.**
  - Designed and developed VCSEL–PD integrated photonic–electronic systems, from epi-wafer, device, to test link, enabling compact high-speed optical paths for random number generation and secure communication.
  - Applied machine learning for device identification, optical-signal classification, and robustness evaluation.
  - Prototyped optical systems demonstrating high-speed multi-key generation and PUF-based authentication.
- **Led development of micro-LED for white light illumination and communication**
  - Developed micro-LEDs-based white-light systems for illumination and optical wireless communication.
  - Conducted optical design, fabrication, and system-level evaluation of micro-LED.
- **Wrote funded proposals, patents, and mentored junior students**

Service and awards

- Served as a reviewer for *IEEE Photonics Technology Letters*, *Photonics Journal*, and *Chinese Optics Letters*
- KAUST Dean's List Award —2023, 2024
- KAUST Career Fair ambassadors —2023
- KAUST "Green Road" excellent volunteer leader —2022
- UESTC First-Class scholarship —2018, 2019, 2020
- National Excellent Youth —2020 (50 awardees nationwide)

Selected Publications and Presentations

- Zhican Zhou*, **Hang Lu***, et al., " Chaos-based scalable optoelectronic physical unclonable functions with AI-driven dynamic authentication." *Nature Electronics*, 2026 (Accepted).
- **Hang Lu,** et al., " Shaping the Light of VCSELs through Cavity Geometry Design." *Light: Science and Applications*, 14. 344 （2025).
- **Hang Lu,** et al., " On-Chip Temper-Resistant Quantum Entropy Source Enabled by Monolithic Emitter and Detector Integration." *International Electron Devices Meeting (IEDM)*, 2025.
- **Hang Lu**, et al., "Low-coherence semiconductor light sources: devices and applications." *npj Nanophotonics* 1.1 (2024): 9.
- **Hang Lu,** et al., "Parallel On-chip Physical Random Number Generator based on Self-chaotic Dynamics of Free-Running Broad-Area VCSEL Array." *IEEE Journal of Selected Topics in Quantum Electronics (**Cover paper**),* 2024.
- **Hang Lu,** et al., "Ultrafast dynamics in tailored VCSEL cavity geometry toward high-efficiency physical random number generation." *APL Photonics* 10.4 (2025).
- **(Invited talk)** Ooi, Boon S., **Hang Lu**, et al., "Design and Applications of Low-Coherence Semiconductor Light Emitters. " in *Conference on Lasers and Electro-Optics (CLEO),* 2024.

Patents

- Ooi, Boon S., Omar Alkhazragi, **Hang Lu**, Nawal Almaymoni, Heming Lin. "A multifunctional monolithically integrated Photodetector-VCSEL chip for massively parallel optical communication, random number generation and quantum key distribution." U.S. Patent Application, 2026 (In progress)
- Two patents related to AR display with Meta. Inc. 2026 (In progress)